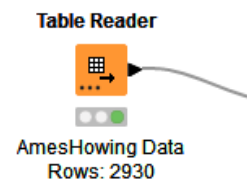


Ames Housing

Regression problem
Outlier removal using Isolation Forests

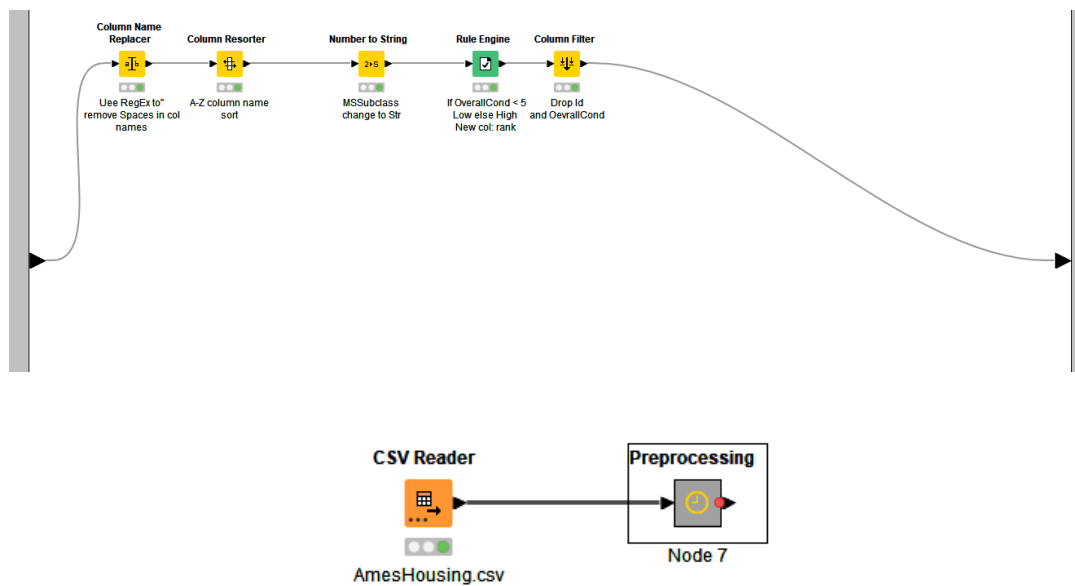
Last amended: 10th Sep, 2026
My folder: C:\Users\ashok\OneDrive\Documents\anomaly_detection\isolationForest
[Reference file](#): 'AmesHousing—numericOutlier' in folder:
C:\Users\ashok\OneDrive\Documents\anomaly_detection\numericOutliers
Dataset: [Kaggle](#)
Virtual Machine:
GitHub: [Clustering/anomaly_detection](#)

In this documentation, we will refer to a [Reference file](#) as mentioned above.
We start from here



Data preprocessing:

Preprocess this dataset as in [this reference file](#) on my machine. Follow Chapter-I of the file to get a final pre-processing metanode.



Chapter-II

Beginning Cross validation

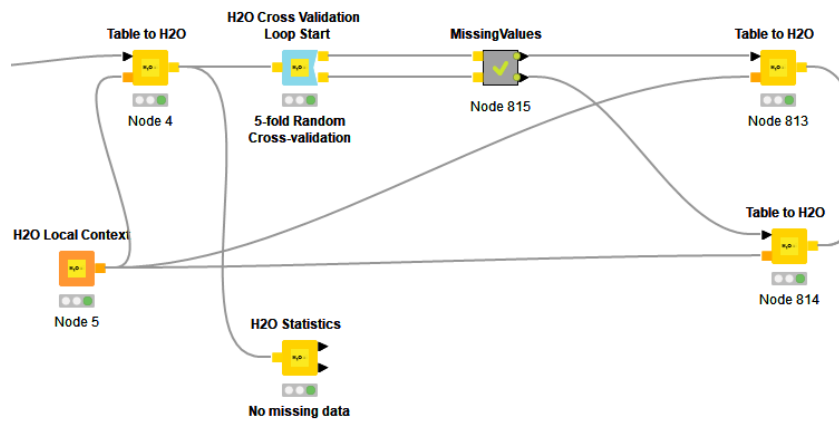


Figure 1: Missing values removal follows reference

Cross validation node: There is nothing special in its configuration. We have five-fold random sampling.

Missing Values Metanode: This node opens up as below:

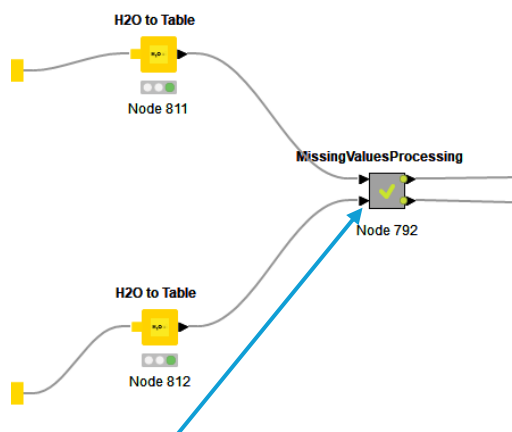
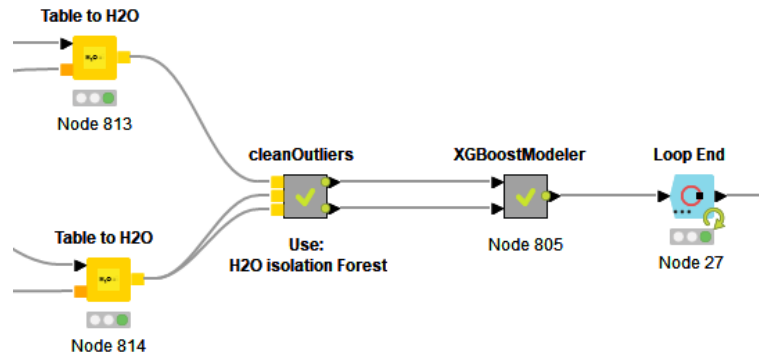


Figure 2: Missing values Processing is exactly the same as in Reference file.

Chapter-III

Cleaning outliers

The flow diagram is as below:



Clean outliers metanode opens as below:

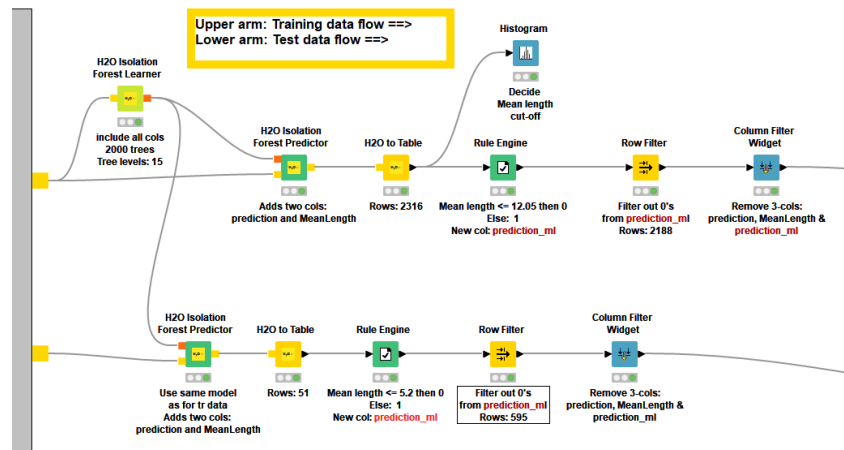


Figure 3: We have one H2o Isolation forest modeler but two predictors--One on each arm.

This flow diagram differs from the Reference file regarding how outliers are removed. In using isolation forests, we have to keep in mind that data is relatively large and we need stable trees.

1. Number of levels (Tree depth): 15
2. Number of models: 2000

The screenshot shows the configuration interface for the H2O Isolation Forest model. Under 'Tree Options', the 'Number of levels (tree depth)' is set to 15. Under 'Forest Options', the 'Number of models' is set to 2,000. The 'Use static random seed' checkbox is checked, with a seed value of 1789369571823. A 'New' button is located next to the seed input field.

Figure 4: Isolation forest settings--I

3. Algorithm Settings (tab) are as below:

Min (weighted) observations: 1

☒ Row sample rate (per tree): 0.8

☐ Row sample size (per tree): 256

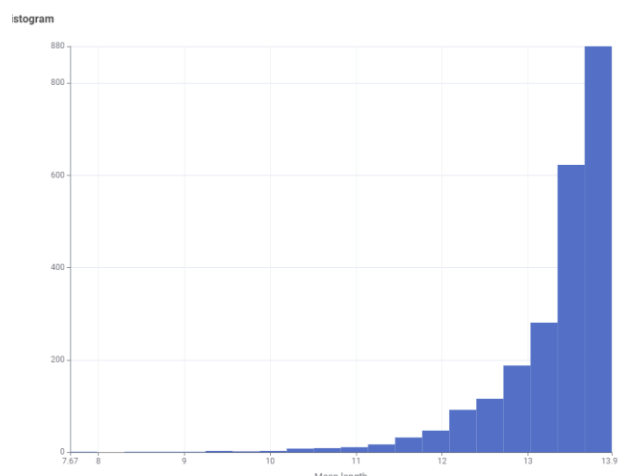
☒ Column sample rate (per tree): 0.8

Relative change of column sample rate per level: 0.9

Figure 5: Isolation Forests Settings—II. Click Algorithm Settings tab and set these parameters to make it more robust.

4. Predictor: No change in settings. Do it as per Reference.

The Mean Length histogram is as below:



Accordingly, our rule engine is:

Expression

```
1 // enter ordered set of rules, e.g.:
2 // $double column name$ > 5.0 => "large"
3 // $string column name$ LIKE "*blue*" => "small and blue"
4 // TRUE => "default outcome"
5 $Mean length$ <= 12.05 => 0
6 TRUE => 1
```

☒ Append Column: prediction_ml

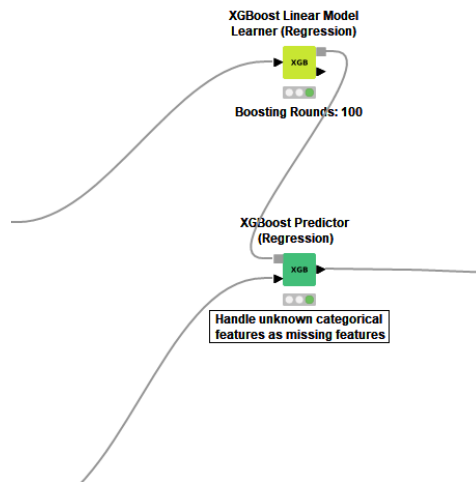
☐ Replace Column: Mean length

Figure 6: Rule Engine settings

Setting in the rest of nodes remain the same as in the Reference document.

Chapter-IV

Xgboost configuration



Configuration for predictor

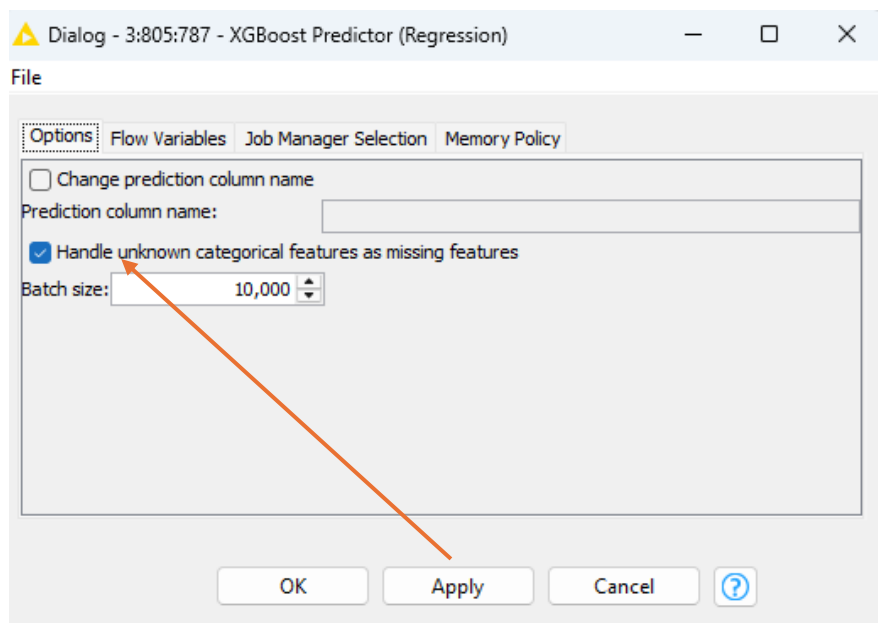


Figure 7: Check mark Handle unknown...

Statistics - 3:790 - Numeric Scorer	
File	
R ² :	0.919
Mean absolute error:	14,812.985
Mean squared error:	443,930,287.45
Root mean squared error:	21,069.653
Mean signed difference:	-136.342
Mean absolute percentage error:	0.091
Adjusted R ² :	0.919

How to experiment:

To experiment, Set Mean Length to 0. This implies that there is no outlier filtering. Next, increase Mean Length. As you increase Mean Length, Mean Absolute Error goes on decreasing till around Mean Length of 12.05. And then goes on decreasing.

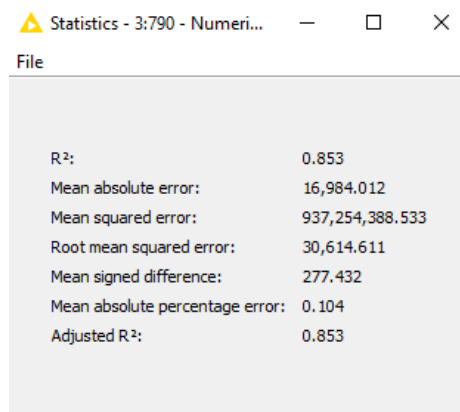
Mean Length < 0.01

Statistics - 3:790 - Numeric Scorer	
File	
R ² :	0.853
Mean absolute error:	16,959.17
Mean squared error:	938,197,243.526
Root mean squared error:	30,630.006
Mean signed difference:	294.52
Mean absolute percentage error:	0.104
Adjusted R ² :	0.853

Mean Length < =2

Statistics - 3:790 - Numeric Scor...	
File	
R ² :	0.853
Mean absolute error:	16,971.318
Mean squared error:	936,783,012.903
Root mean squared error:	30,606.911
Mean signed difference:	287.075
Mean absolute percentage error:	0.104
Adjusted R ² :	0.853

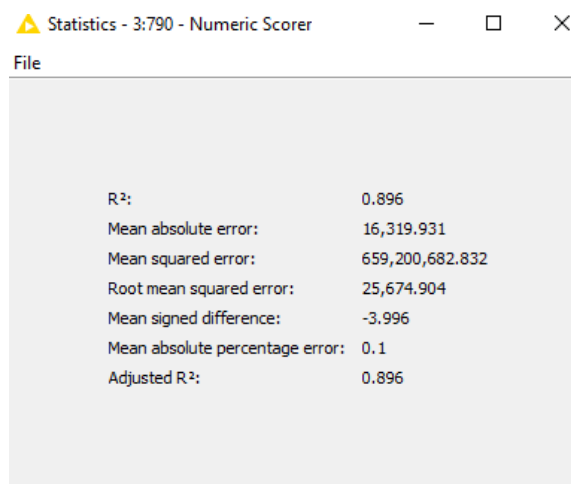
Mean Length ≤ 5



A screenshot of a software window titled "Statistics - 3:790 - Numeri...". It displays a table of statistical metrics for a model where Mean Length is less than or equal to 5. The metrics include R-squared, Mean absolute error, Mean squared error, Root mean squared error, Mean signed difference, Mean absolute percentage error, and Adjusted R-squared.

R ² :	0.853
Mean absolute error:	16,984.012
Mean squared error:	937,254,388.533
Root mean squared error:	30,614.611
Mean signed difference:	277.432
Mean absolute percentage error:	0.104
Adjusted R ² :	0.853

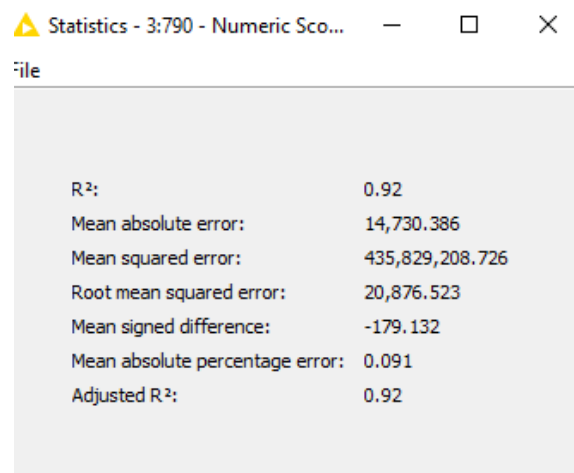
Mean Length ≤ 10



A screenshot of a software window titled "Statistics - 3:790 - Numeric Scorer". It displays a table of statistical metrics for a model where Mean Length is less than or equal to 10. The metrics include R-squared, Mean absolute error, Mean squared error, Root mean squared error, Mean signed difference, Mean absolute percentage error, and Adjusted R-squared.

R ² :	0.896
Mean absolute error:	16,319.931
Mean squared error:	659,200,682.832
Root mean squared error:	25,674.904
Mean signed difference:	-3.996
Mean absolute percentage error:	0.1
Adjusted R ² :	0.896

Mean Length ≤ 12



A screenshot of a software window titled "Statistics - 3:790 - Numeric Sco...". It displays a table of statistical metrics for a model where Mean Length is less than or equal to 12. The metrics include R-squared, Mean absolute error, Mean squared error, Root mean squared error, Mean signed difference, Mean absolute percentage error, and Adjusted R-squared.

R ² :	0.92
Mean absolute error:	14,730.386
Mean squared error:	435,829,208.726
Root mean squared error:	20,876.523
Mean signed difference:	-179.132
Mean absolute percentage error:	0.091
Adjusted R ² :	0.92

Mean Length < 13

Statistics - 3:790 - Numeric Scorer

file

R ² :	0.922
Mean absolute error:	12,930.327
Mean squared error:	333,041,526.634
Root mean squared error:	18,249.425
Mean signed difference:	-129.252
Mean absolute percentage error:	0.077
Adjusted R ² :	0.922

Mean Length < 14

No results. Empty table.

#####